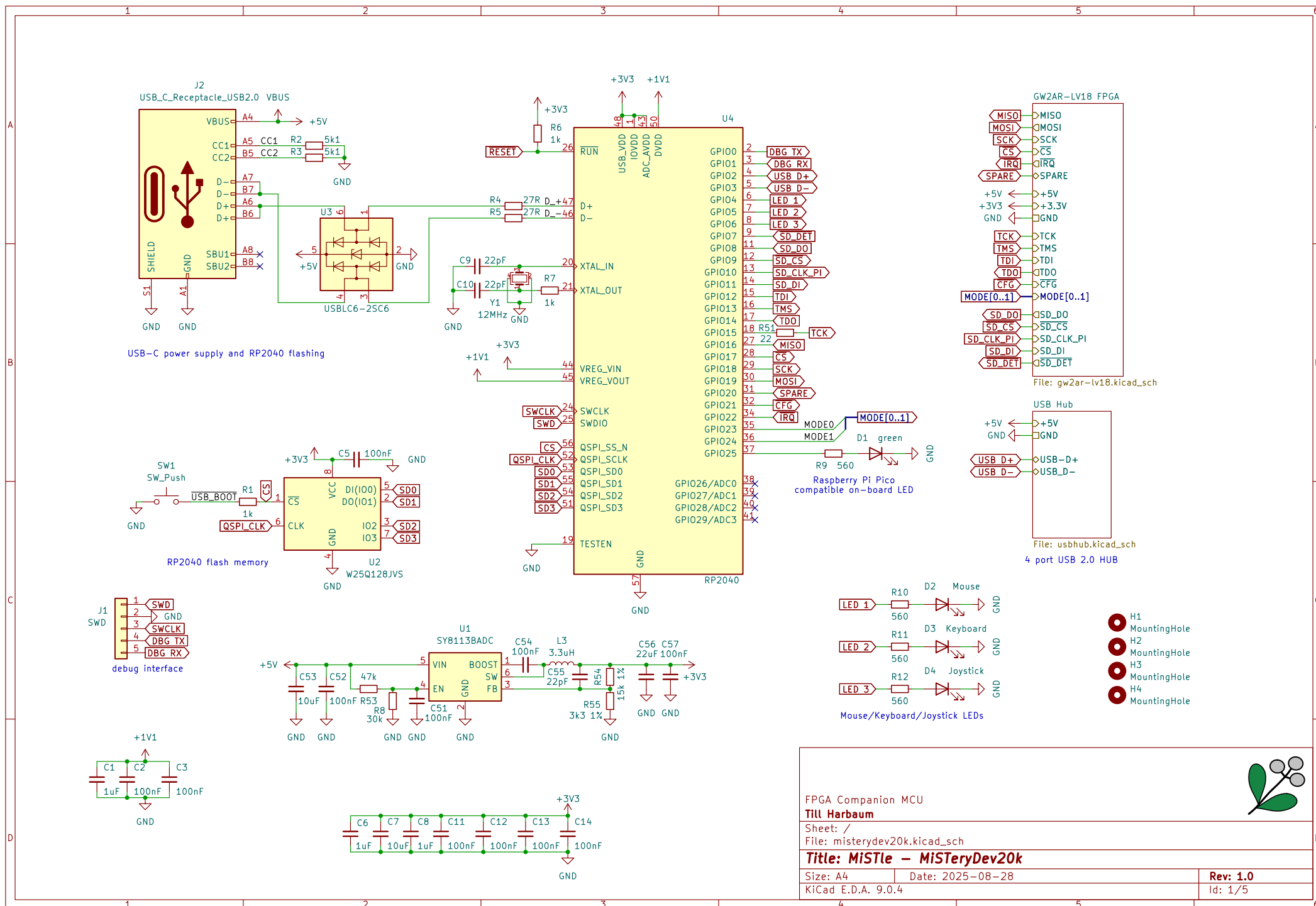
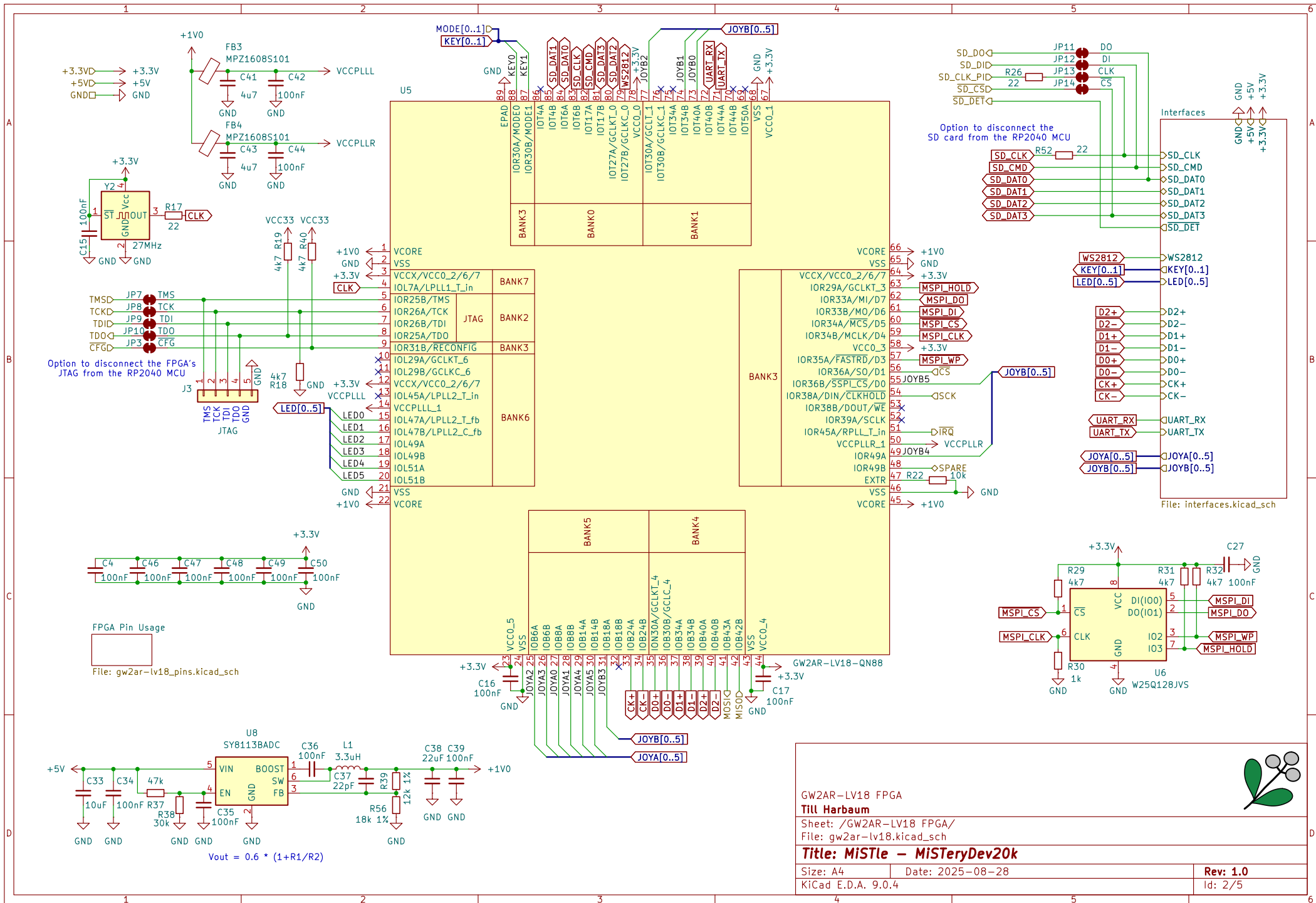






GW2AR-LV18 FPGA

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Sheet: /GW2AR-LV18 FPGA/

File: gw2ar-lv18.kicad_sch

Title: MiSTle - MiSTeryDev20k

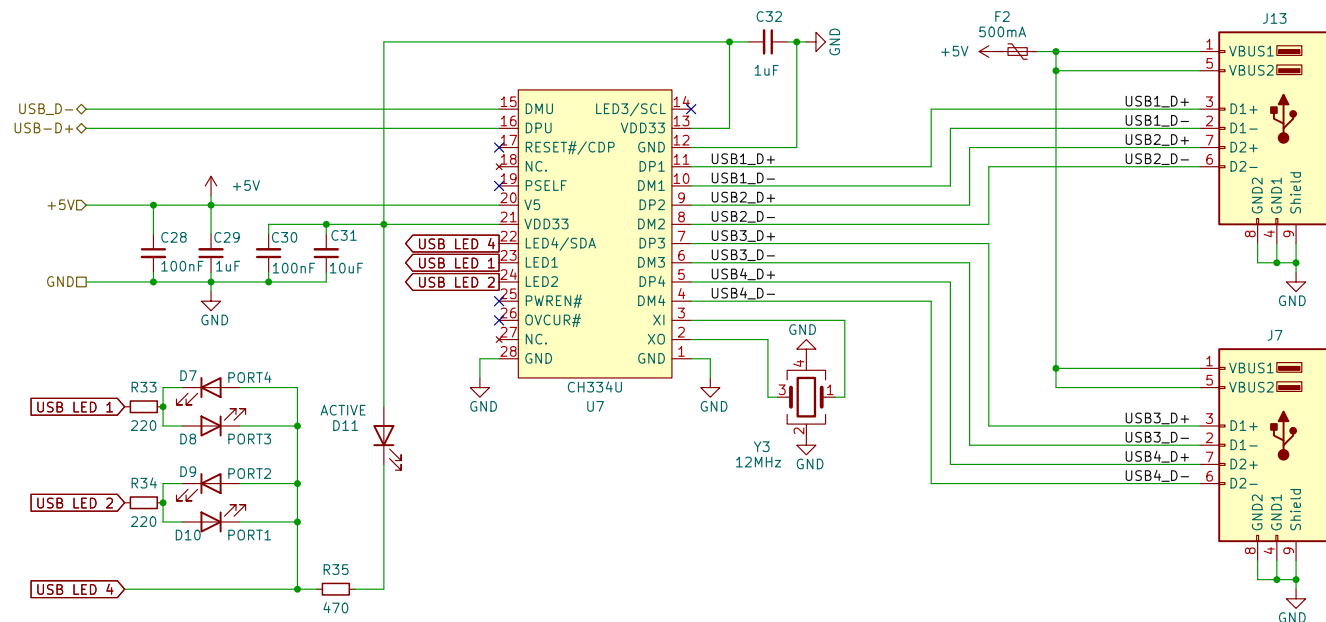
Size: A4 Date: 2025-08-28

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USB Hub

Stefan Voss, Till Harbaum

Sheet: /USB Hub/

File: usbbhub.kicad_sch

Title: MiSTle – MiSTeryDev20k

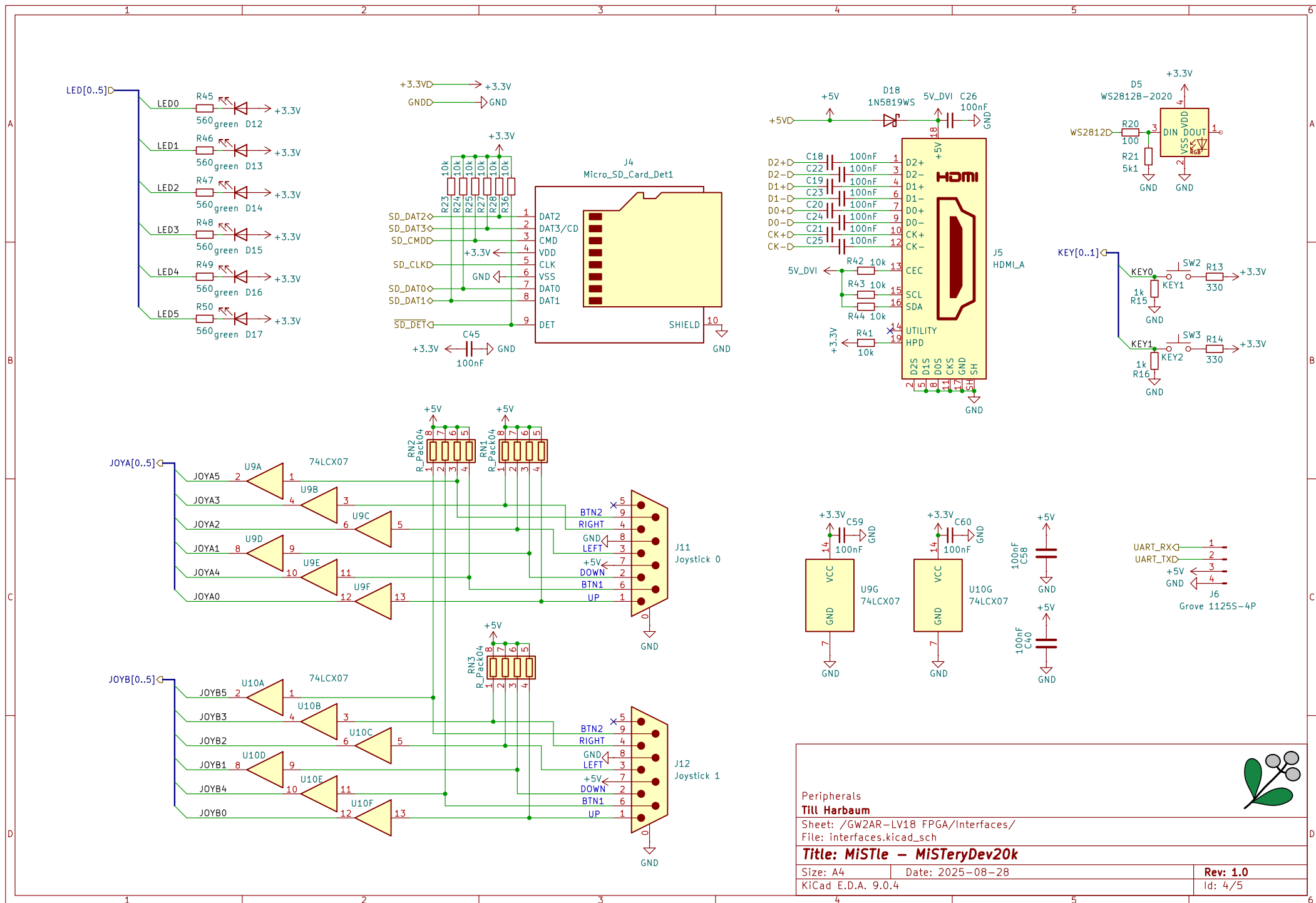
Size: A4 Date: 2025-08-28

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Peripherals
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Sheet: /GW2AR-LV18 FPGA/Interfaces/
File: interfaces.kicad_sch

Title: MiSTle - MiSTeryDev20k

Size: A4 Date: 2025-08-28

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Pin	Function	TN20k (yellow=on hdr)	NanoMig	Pin	Function	TN20k (yellow=on hdr)	NanoMig
1	Vcore			45	Vcore		
2	Vss			46	Vss		
3	Vccc			47	EXTR		
4	IOL7A	SYS_CLK	CLOCK	48	IOR49B	LCD_DE	M0S5
5	IOR25B	SYS_TMS		49	IOR49A	LCD_BL	SPARE4
6	IOR26A	SYS_TCK		50	Vccpll		
7	IOR26B	SYS_TDI		51	IOR45A	PA_EN	M0S4
8	IOR25A	SYS_TDO		52	IOR39A	EDID_DAT	
9	IOR31B	RECFG		53	IOR38B	EDID_CLK	
10	IOL29A	5351_CKP		54	IOR38A	I2S_DIN	M0S3
11	IOL29B	5351_CKN		55	IOR36B	I2S_LRCK	SPARE5
12	Vccx			56	IOR36A	I2S_BCLK	M0S2
13	IOL45A	HSPI_SCLK	BL616_SCLK	57	IOR35A	MSPI_WP	MSPI_WP
14	Vccpll			58	Vcc		
15	IOL47A	LED0	LED0	59	IOR34B	MSPI_CLK	MSPI_CLK
16	IOL27B	LED1	LED1	60	IOR34A	MSPI_CS	MSPI_CS
17	IOL49A	LED2	LED2	61	IOR33B	MSPI_DI	MSPI_DI
18	IOL49B	LED3	LED3	62	IOR33A	MSPI_DO	MSPI_DO
19	IOL51A	LED4	LED4	63	IOR29A	MSPI_HOLD	MSPI_HOLD
20	IOL51B	LED5	LED5	64	Vccx		
21	Vss			65	Vss		
22	Vcore			66	Vcore		
23	Vcc			67	Vcc		
24	Vss			68	Vss		
25	IQB6A	LCD_HS	IO2	69	IOT50A	SYS_TX	
26	IQB6B	LCD_VS	IO3	70	IOT44B	SYS_RX	
27	IQB8A	LCD_B7	IO0	71	IOT44A	HSPI_DIN0	MIDI_OUT
28	IQB8B	LCD_B6	IO1	72	IOT40B	HSPI_DIN1	MIDI_IN
29	IQB14A	LCD_B5	IO4	73	IOT40A	HSPI_DIN2	SPARE0
30	IQB14B	LCD_B4	IO5	74	IOT34B	HSPI_DIN3	SPARE1
31	IQB18A	LCD_B3	SPARE3	75	IOT34A	HSPI_DIR	BL616_DIR
32	IQB18B	LCD_G7		76	IOT30B	HSPI_DAT	BL616_DAT
33	IQB24A	DVI_CKP	DVI_CKP	77	IOT30A	LCD_CLK	SPARE2
34	IQB24B	DVI_CKN	DVI_CKN	78	Vcc		
35	IQB30A	DVI_D0P	DVI_D0P	79	IOT27B	WS2812	WS2812
36	IQB30B	DVI_D0N	DVI_D0N	80	IOT27A	SDIO_D2	SDIO_D2
37	IQB34A	DVI_D1P	DVI_D1P	81	IOT17B	SDIO_D3	SDIO_D3
38	IQB34B	DVI_D1N	DVI_D1N	82	IOT17A	SDIO_CMD	SDIO_CMD
39	IQB40A	DVI_D2P	DVI_D2P	83	IOT6B	SDIO_CLK	SDIO_CLK
40	IQB40B	DVI_D2N	DVI_D2N	84	IOT6A	SDIO_D0	SDIO_D0
41	IQB43A	LCD_R4	M0S0	85	IOT4B	SDIO_D1	SDIO_D1
42	IQB42B	LCD_R3	M0S1	86	IOT4A	HSPI_CS	BL616_CS
43	Vss			87	IOR20B	KEY2	BTN_RESET
44	Vcc			88	IOR30A	KEY1	BTN_USER

FPGA pin usage

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Sheet: /GW2AR-LV18 FPGA/FPGA Pin Usage/

File: gw2ar-lv18_pins.kicad_sch

Title: MiSTle – MiSTeryDev20k

Size: A4

Date: 2025-08-28

Rev: 1.0

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